University of Milano-Bicocca
Department of Statistics and Quantitative Methods
Master's Programme — Data Science

Financial Vulnerability in Italy: Who Is Left Behind in Financial Literacy?

An Individual-Level Analysis of the Bank of Italy IACOFI 2023 Survey

Amin Entezari **Ali Sedghiye**

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Keywords. Financial literacy; financial vulnerability; OECD/INFE composite score; machine learning; Italy.

Dataset. IACOFI 2023 — *Indagine sulle Competenze Finanziarie degli Italiani* (Bank of Italy Survey on Financial Literacy and Digital Financial Skills of Italian Adults).

Declaration on the use of AI tools. AI-based assistants (a large language model) were used as a *support* instrument: to help debug and structure the Python code, to help organise the report, and to refine the English prose. The research question, all analytical decisions, the interpretation of the results and the conclusions were defined and verified by the authors. Every reported number was checked directly against the data; nothing was reported without verification.

Abstract

Financial literacy — the capacity to understand and act on financial information — is a key determinant of individual economic well-being, yet it is unevenly distributed across the population. Using the Bank of Italy IACOFI 2023 survey ($n = 4{,}862$ adults), this study asks *which demographic, socioeconomic and behavioural factors predict financial vulnerability among Italian adults, and which groups are most at risk.* A composite literacy score (0–21) was constructed following the OECD/INFE methodology, combining knowledge, behaviour and attitudes, and validated for internal consistency. The analysis proceeds through five stages: score construction, exploratory data analysis with hypothesis testing, supervised classification, behavioural feature engineering, and unsupervised clustering. We find that vulnerability is concentrated among the unemployed, the young (18–29), and the low-income and low-education groups, with all gaps statistically significant ($p < 0.001$). A demographics-only model predicts vulnerability weakly ($\approx 55\%$ accuracy); adding engineered behavioural features (financial resilience, budgeting, product holdings) raises accuracy to $\approx 70\%$, and the financial-resilience index emerges as the single strongest predictor — ahead of every demographic variable. K-Means clustering isolates a distinct worst-off group: elderly, low-income, digitally-excluded individuals. The central message for policymakers and financial-education providers is that vulnerability is driven more by what people *do*, and the buffers they hold, than by who they *are*, implying that behaviour-focused and life-stage-targeted interventions should be more effective than purely demographic targeting.

1 Research Question

1.1 Statement and definitions

The central question of this study is:

> *Which demographic, socioeconomic and behavioural factors predict financial vulnerability among Italian adults, and which population groups are most at risk?*

Before the analysis can proceed, the key concepts must be defined precisely, because the validity of the whole study depends on how these abstract notions are turned into measurable quantities.

Financial literacy. Following the OECD International Network on Financial Education (INFE) framework, financial literacy is not a single skill but the combination of three distinct dimensions. *Knowledge* is the cognitive understanding of fundamental financial concepts — inflation, interest, risk and diversification. *Behaviour* is the set of concrete actions a person takes — saving regularly, budgeting, paying bills on time, planning ahead. *Attitudes* are the underlying dispositions — whether a person is oriented towards the long-term financial future or towards immediate consumption. A person can be strong in one dimension and weak in another; this is why all three must be measured.

Financial literacy score. We turn these three dimensions into a single composite indicator ranging from 0 to 21 (the sum of a 0–7 knowledge component, a 0–9 behaviour component and a 0–5 attitudes component). The construction is detailed in Section 4.1. This composite is the *target variable* of the study.

Financial vulnerability. We operationalise vulnerability as scoring *below the population median* of the literacy score. This is a relative, population-anchored definition: rather than fixing an arbitrary absolute threshold, we identify the half of the population that is comparatively worse-off. This choice makes the target balanced and interpretable.

The question is deliberately framed as one of *problem identification*. The objective is not to build the most accurate possible prediction machine, but to locate the groups that are worst-off and to understand the factors that drive their disadvantage. The predictive models that follow are used as instruments to confirm and quantify those factors, not as the goal in themselves.

1.2 Motivation and stakeholders

Financial literacy matters because it shapes some of the most consequential decisions in a person's life: how much to save, how to manage debt, whether to plan for retirement, and how to avoid fraud and scams. Empirical research (e.g. Lusardi and Mitchell, 2014) links low financial literacy to poorer financial outcomes and greater fragility in the face of economic shocks. Identifying *who* is most vulnerable, and *why*, is therefore of direct interest to several stakeholders:

- **Public policymakers** — in particular the Bank of Italy and the national committee for financial education (Comitato Edufin), which design national financial-education strategies and must decide which groups to target and through which channels.
- **Financial-education providers, schools and NGOs** — which deliver programmes on the ground and need evidence on which groups and which behaviours to prioritise.
- **Financial institutions** — which have a commercial and prudential interest in a customer base that is financially capable and resilient.

The study is intentionally scoped to a single survey wave and a single, focused question. This follows the spirit of the course: a relevant question, soundly addressed, is more valuable than an exhaustive but shallow treatment of the topic.

2 The Data

2.1 Description

The dataset is the **IACOFI 2023** survey, conducted by the Bank of Italy following the OECD/INFE 2022 Toolkit for Measuring Financial Literacy and Financial Inclusion. The survey covers **4,862 individuals** aged 18–79, drawn to be representative of the Italian adult population, and was administered by Computer-Assisted Telephone Interviewing (CATI) in February and March 2023. The 2023 edition is the most recent and the richest of the three IACOFI waves (2017, 2020, 2023), and is the first to include questions on digital financial skills.

The raw file contains **219 variables**, which we group into five thematic blocks:

- **Demographics and socioeconomics:** gender (QD1), age (QD7), macro-region (QD2: North-West, North-East, Centre, South, Islands), type of municipality (QD3), household composition (QD5), work status (QD10), education (QD9), income band (QD13), nationality (QD12), internet access (QD14).
- **Financial knowledge (QK):** questions on inflation and purchasing power, simple and compound interest, the risk–return relationship, and diversification.
- **Financial behaviour (QF, QS2):** budgeting practices, active saving, the ability to absorb a financial shock, and timely bill payment.
- **Financial attitudes (QS1, QS3):** the balance between saving and spending, satisfaction with one's financial situation, and the sense of control over one's finances.
- **Digital financial skills (QS4, QP8, QP9):** new in 2023, covering online financial safety, trust in FinTech, and digital financial activity.

The dataset also contains a sampling weight (`wght`), which restores representativeness of the Italian population when applied. As an external enrichment, we attached two official ISTAT 2023 macro-regional indicators — the unemployment rate and the share of households with internet access — merged onto the five macro-regions to provide structural economic context for the geographic dimension of the analysis.

2.2 Strengths and weaknesses with respect to the goal

A central principle of the course is that a data-science result is only as valid as the data and the process behind it. We therefore assess the data quality explicitly, in the spirit of the ISO 25012 dimensions (accuracy, completeness, consistency, credibility).

Strengths. The data are *credible and accurate*: they are collected professionally by a central bank using a standardised international methodology, which also makes results comparable across countries. They are *rich and complete*: all three OECD literacy dimensions are present, alongside behavioural, attitudinal and digital items, so the target can be constructed faithfully. They are *representative*: the sampling design plus weights support inference to the national population. And, crucially for our question, they are *individual-level*: unlike country-level indicators, microdata allow us to study heterogeneity *within* the population — which is exactly where vulnerability hides.

Weaknesses. First, the income variable (QD13) has *31% missing or refused values*. Income non-response is rarely random; it tends to concentrate at the two extremes of the distribution (the very poor and the very rich), which can bias any analysis that conditions on income. We treat this transparently rather than imputing aggressively. Second, the survey is *cross-sectional*: a single snapshot in time. This prevents causal inference — we can observe that low income and low literacy go together, but we cannot establish which causes which. Third, the behavioural and attitudinal items are *self-reported*, and may be affected by social-desirability bias (respondents over-stating prudent behaviour). Fourth, geography is *coarse*: only five macro-regions are available, so the ISTAT regional enrichment, while informative as context, carries little statistical power on its own (it adds only five distinct values). These limitations are acknowledged throughout and revisited in the conclusion.

3 Methodology

3.1 The overall logic thread

The study is built as a single coherent argument, advancing in five stages. Each stage answers a sub-question and feeds the next.

1. **Construct and validate the target.** We first build the OECD/INFE composite literacy score from the raw survey items, and check that the three dimensions are internally consistent — i.e. that the composite is meaningful and not redundant. Without a valid target, everything downstream would be invalid.
2. **Locate the gaps (exploratory data analysis).** We describe how the literacy score varies across demographic and socioeconomic groups, and we test each apparent gap for statistical significance, so that we do not build a story on sampling noise.
3. **Quantify the predictors (supervised learning).** We train classifiers to predict vulnerability from demographics, and we read their feature importances to obtain a data-driven ranking of which factors matter most.
4. **Enrich with behaviour (feature engineering).** We construct new behavioural indicators from the raw survey and test whether they improve predictive power and change the ranking of drivers. This is the methodological heart of the study.
5. **Discover hidden profiles (unsupervised learning).** Finally, we cluster respondents to see whether the data themselves reveal distinct, interpretable vulnerable segments that the variable-by-variable analysis might miss.

Throughout, the machine-learning models are treated as *instruments of confirmation and quantification*, not as the objective. This reflects the course's warning against "purely technical" reports that merely race algorithms against one another.

3.2 The tools, and why

All analysis was carried out in **Python** inside a reproducible **Docker**/Jupyter environment, organised into five sequential notebooks (Appendix). The main libraries were `pandas` and `numpy` (data handling), `matplotlib` and `seaborn` (visualisation), `scikit-learn` (machine learning) and `scipy` (statistical tests).

Composite score — OECD/INFE method. We use the international standard because it is the very methodology the Bank of Italy adopts, which guarantees that our target is valid and comparable, rather than an arbitrary construction of our own.

Statistical tests — t-tests and ANOVA. A difference in group means observed in a sample may or may not reflect a real difference in the population. The independent-samples t-test (for two groups) and one-way ANOVA (for three or more) quantify the probability that an observed gap could arise by chance. We require $p < 0.05$ to call a gap significant.

Logistic Regression. A transparent linear classifier, used as the baseline. It estimates the probability of vulnerability as a function of the features and is easy to interpret.

Random Forest and Gradient Boosting. Two ensemble methods that combine many decision trees. They capture non-linearities and interactions that a linear model misses, and — importantly for our problem-identification goal — they output *feature importances* that rank the predictors. Gradient Boosting builds trees sequentially, each correcting the errors of the previous, and was retained as the best performer.

K-Means clustering. An unsupervised method that partitions respondents into k groups by minimising within-group distance. It is simple and interpretable; we choose k with the elbow method (Section 4.6).

Data preparation. The survey encodes non-responses with negative sentinel codes: -97 ("don't know"), -98 ("not applicable"), -99 ("refused") and -999 ("irrelevant answer"). A crucial cleaning step — and a real validity issue — is that these must be converted to genuine missing values *before* any computation; treating -99 as a numeric score would badly distort every average and model. After recoding, features were standardised where the method required it (logistic regression, K-Means), and the supervised models used a stratified 80/20 train–test split with 5-fold cross-validation to obtain honest performance estimates.

4 Results

4.1 Stage 1 — Constructing and validating the score

Each of the three dimensions is scored separately and then summed. *Knowledge* (0–7) awards one point for each correctly answered financial-knowledge question (inflation, interest paid, simple interest, compound interest, risk–return, the meaning of inflation, and diversification). *Behaviour* (0–9) awards points for prudent practices (making a budget plan, having actively saved, being able to absorb a shock, paying bills on time, considering affordability before buying, having money left at month's end, setting long-term goals, watching one's finances, and not preferring spending over saving). *Attitudes* (0–5) awards points for a future-oriented disposition (disagreeing that "money is there to be spent", satisfaction with one's finances, setting long-term goals, not worrying constantly about expenses, and feeling in control of one's finances). For each respondent, questions left unanswered were excluded from that person's denominator, so that missing answers neither count as correct nor artificially deflate the score.

The resulting national mean is **11.70 out of 21** (standard deviation 3.18), a value consistent with the OECD's published benchmark for Italy, and the distribution is approximately bell-shaped (Figure 1). As a robustness check on the sampling design, we recomputed the mean using the official survey weights: the weighted mean is 11.52, just 0.18 points below the unweighted figure. This negligible difference tells us that the unweighted analyses — which we use throughout for simplicity — are reliable and not distorted by the sampling scheme.

A composite indicator is only meaningful if its components belong together but are not mere duplicates of one another. We therefore examined the correlations between the three sub-scores (Table 1). They are low to moderate: Knowledge–Behaviour = 0.15, Behaviour–Attitudes = 0.28, and Knowledge–Attitudes $\approx$ 0.00. This is exactly the desirable pattern: the dimensions are related enough to form a coherent construct, yet distinct enough that each contributes its own information. The composite is therefore valid and not redundant — a point that matters because, as the course stresses, an indicator one does not understand and validate can be deeply misleading.

Table 1. Composite-score components and internal-consistency check.

Dimension	Max	Mean	SD	Note
Knowledge	7	5.02	1.56	relatively strong
Behaviour	9	4.81	1.98	moderate
Attitudes	5	1.77	1.31	weakest dimension
Composite	**21**	**11.70**	**3.18**	target variable

Inter-dimension correlations: Knowledge–Behaviour = 0.15*; Behaviour–Attitudes* = 0.28*; Knowledge–Attitudes* ≈ 0.00.

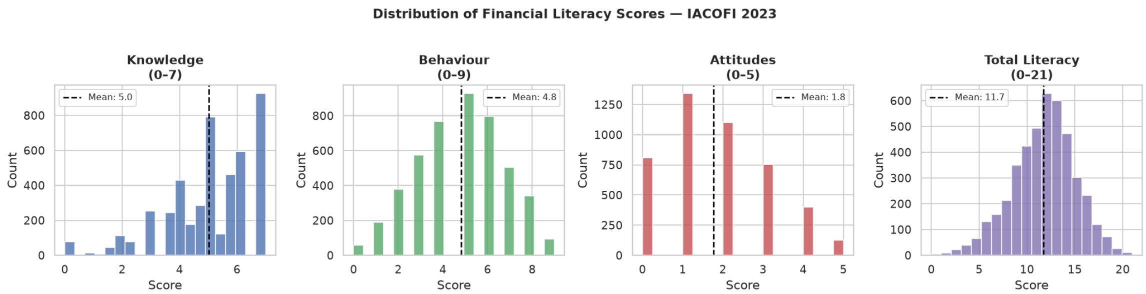


Figure 1. Distribution of the three sub-scores and the composite financial literacy score (IACOFI 2023). Knowledge is left-skewed (most people answer several questions correctly), attitudes cluster at low values, and the composite is approximately symmetric around its mean of 11.70.

4.2 Stage 2 — Locating the gaps

With a valid target in hand, we examined how literacy varies across the population. Table 2 summarises the mean score within the key groups, and the patterns are striking.

The single largest raw gap is by **work status**. The unemployed score 9.04 on average — the lowest of any group, and almost three points below the national mean — while the self-employed reach 12.29. The intuition is that economic exclusion and low literacy reinforce one another: those without stable employment have fewer financial decisions to practise on and fewer resources to build their competence.

There is also a clear and monotone **income gradient**: scores rise from 11.39 in the lowest income band, to 11.97 in the middle, to 13.01 in the highest. Higher earners hold more financial products and face more financial decisions, which both reflects and reinforces literacy.

The most surprising finding is the **age pattern**. One might expect young people — digitally fluent and recently educated — to be the most literate. Instead, the 18–29 cohort scores *lowest of all* (10.25), below even the 70–79 group (Figure 2a). We call this the "youth paradox". The most plausible explanation is a *financial-experience gap*: literacy is built largely through living with real financial responsibilities — a mortgage, a pension, a household budget — which young adults have not yet accumulated, regardless of their schooling or digital skills.

Finally, **education and income compound** one another (Figure 2b). The heatmap of mean score by education × income shows a clear diagonal: the worst-off cell — middle-school education *and* low income — scores 11.05, while the best-off cell — university education *and* high income — reaches 13.45, a gap of 2.40 points. Disadvantage is therefore not additive but cumulative: being low on two dimensions at once is worse than the sum of being low on each separately.

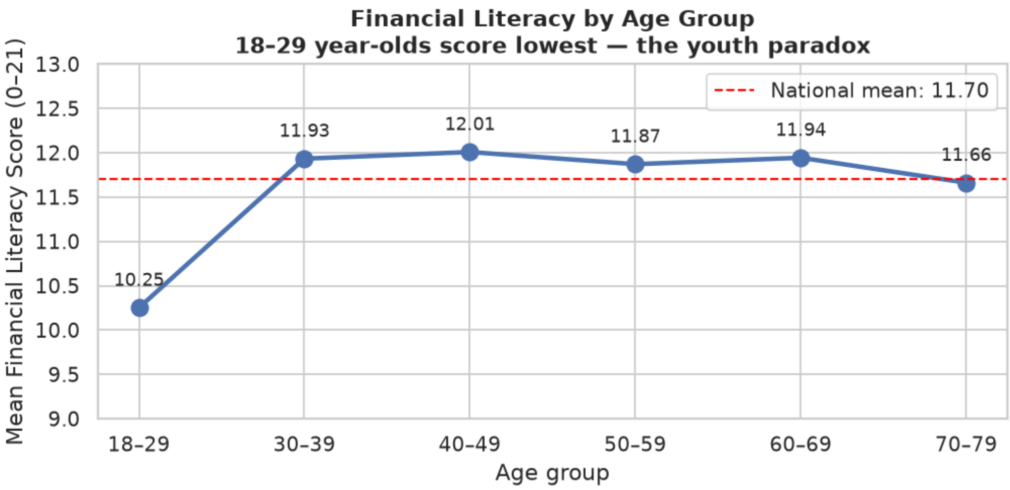
Table 2. Mean financial literacy score by selected groups (national mean = 11.70).

Dimension	**Group**	**Mean (0–21)**
Work status	Unemployed	9.04
	Employed	11.91
	Self-employed	12.29
Income band	Low ($\leq$ €1,750)	11.39
	Mid (€1,751–2,900)	11.97
	High ($>$ €2,900)	13.01
Age	18–29 (lowest)	10.25
	40–49 (highest)	12.01
Education	Middle school or below	11.35
	High school	11.52
	University or above	12.26

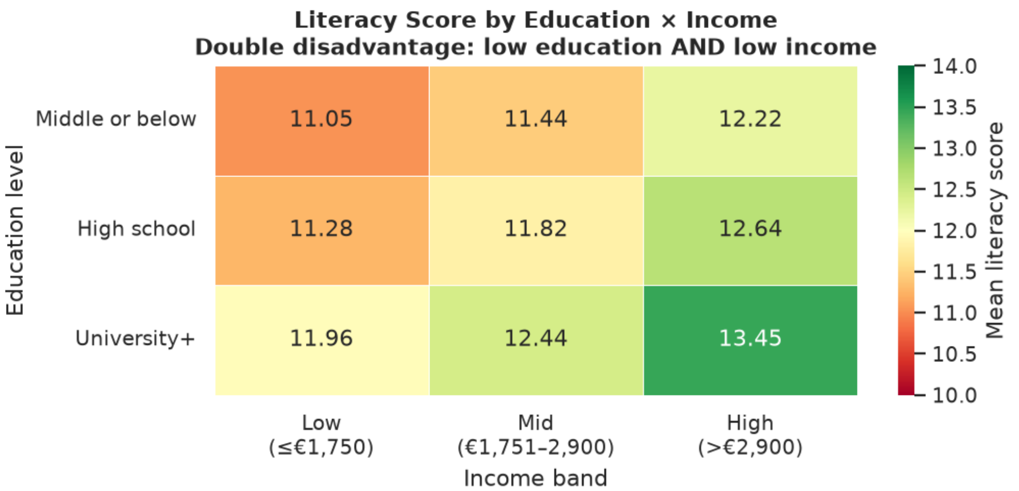
To make sure these gaps are real and not artefacts of sampling, every one of them was tested formally (Table 3). The differences are highly significant in all cases, with p-values far below 0.001 and large test statistics. We can therefore state with confidence that the gaps reflect genuine population differences, not random variation in the sample.

Table 3. Significance tests of the main group differences.

Comparison	**Statistic**	**p-value**
Unemployed vs Employed	$t = -8.59$	< 0.001
Youth (18–29) vs Mid (40–49)	$t = -9.23$	< 0.001
Low vs High income	$t = -8.51$	< 0.001
Education (three groups, ANOVA)	$F = 26.43$	< 0.001



(a) The youth paradox: the 18–29 group scores lowest of all age cohorts.



(b) Education × income: the cumulative "double disadvantage".

Figure 2. Two key exploratory patterns underlying the analysis.

4.3 Stage 3 — Predicting vulnerability from demographics

We then asked how well vulnerability (a below-median score) can be predicted from the eight demographic and socioeconomic features alone. Two classifiers were trained: a Logistic Regression baseline and a Random Forest. Both achieved only *modest* accuracy — 58.2% and 54.9% respectively under 5-fold cross-validation, against a 50% baseline for a balanced two-class problem.

This modest performance is not a failure of the code; it is a substantive finding. Demographic variables are *distal* causes of literacy: age, region or income shape the conditions in which literacy develops, but they do not directly determine it. Two people identical in age, education and income can differ greatly in literacy depending on factors the survey does not record — how their parents handled money, whether they have lived through a financial crisis, their personality. A ceiling around 55% is therefore exactly what theory would predict.

The Random Forest's feature importances nonetheless gave a useful, data-driven ranking, placing *age* first, then *education* and *work status*, with gender and internet access least important. Reassuringly, this matches the ranking suggested by the exploratory analysis, providing convergent evidence.

4.4 Stage 4 — Feature engineering: behaviour beats demographics

The key methodological step was to move beyond demographics and construct features that capture what people actually *do* financially. From the raw survey we engineered five behavioural indicators:

- **Financial resilience index** — combining whether the respondent can absorb a sudden major expense (QF4), whether they have money left at month's end (QS2_4), and how long they could survive without income (QF13).
- **Financial product breadth** — the number of financial products currently held.
- **Budgeting-behaviour score** — the number of budgeting practices the respondent reports (QF2).
- **Active-saver flag** — whether the respondent saved in any form over the past year.
- **Self-rated knowledge** — the respondent's own assessment of their financial knowledge (reverse-coded).

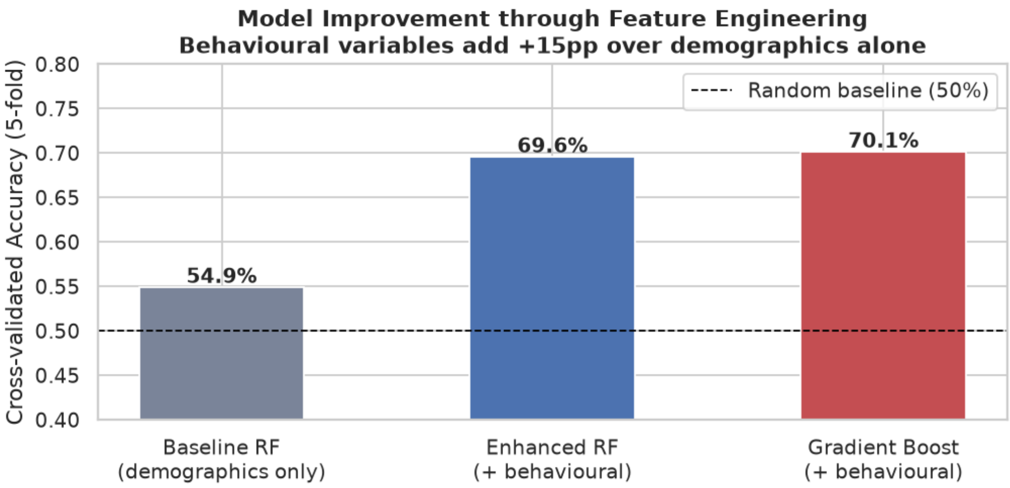
The correlations of these new features with the literacy score (Table 4, lower panel) are markedly higher than those of any demographic variable: the resilience index alone correlates at 0.44, against roughly 0.10–0.15 for the strongest demographics. Re-training the models with the expanded feature set lifted cross-validated accuracy from 54.9% to **70.1%** (Gradient Boosting) — an improvement of about fifteen percentage points (Figure 3a). And the ranking of drivers changed decisively: the *resilience index became the single most important predictor* in the final model, ahead of age and every other demographic feature (Figure 3b).

This step also illustrates one of the course's core concepts, the **bias–variance trade-off**. The demographic baseline is a *high-bias, low-variance* model: it is too simple to capture a complex human trait, so it under-fits and its accuracy is low but stable. Adding the behavioural features — which are *proximal* to literacy — reduces bias substantially (accuracy rises from 55% to 70%) while variance remains low (the cross-validation standard deviation stays around ±1.3%). The gain therefore comes from genuinely more informative inputs, not from over-fitting; an over-fitted model would have shown high variance and a large train–test gap, which we do not observe.

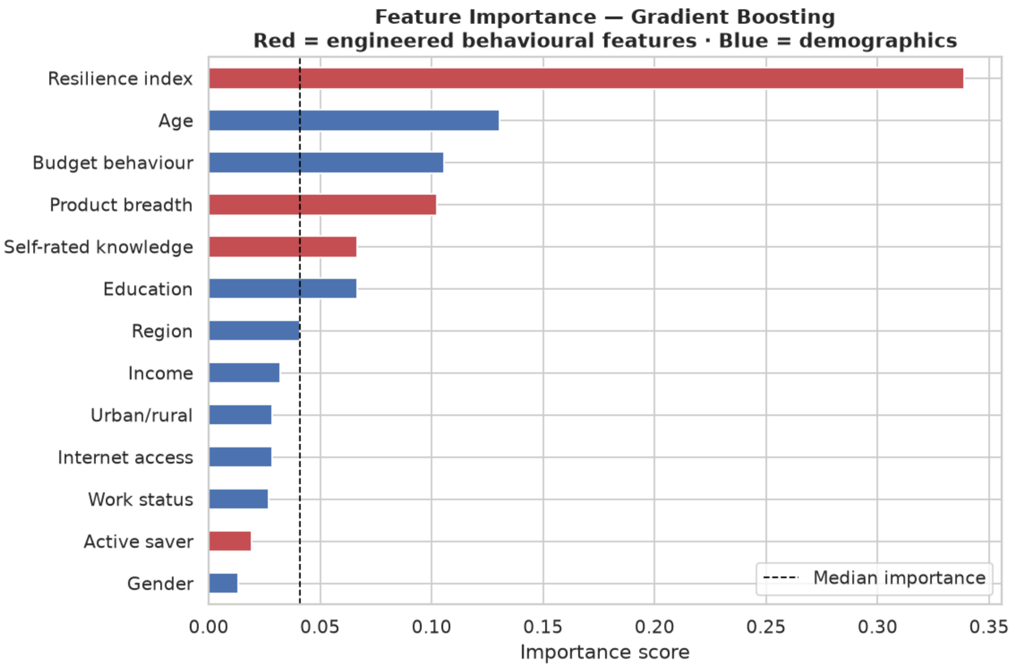
Table 4. Model comparison and the most informative engineered features.

Model	Features	CV accuracy
Logistic Regression	8 demographic	58.2%
Random Forest	8 demographic	54.9%
Random Forest	13 (+ behavioural)	69.6%
Gradient Boosting	**13 (+ behavioural)**	**70.1%**

Correlation of engineered features with the literacy score: resilience index 0.44*; product breadth* 0.37*; budgeting* 0.32*; active saver* 0.27*; self-rated knowledge* 0.20.



(a) Accuracy gain: +15 percentage points from behavioural features.



(b) Final-model importances; behavioural features (red) dominate.

Figure 3. The final Gradient Boosting model: behaviour out-predicts demographics, led by the resilience index.

4.5 Stage 5 — Unsupervised profiles: the elderly offline group

The variable-by-variable analysis can miss combinations of characteristics that define a coherent group. To uncover such hidden profiles, we applied K-Means clustering to the demographic features. The number of clusters was chosen with the *elbow method*: plotting the within-cluster dispersion (inertia) against k and looking for the point where adding clusters stops yielding large gains, which suggested $k = 4$.

The clustering isolated a distinct **worst-off segment** of 261 individuals: average age around 69, low income, and — defining the group — *no internet access at all*, with the lowest mean literacy of any cluster (11.02). This group is doubly disadvantaged: it is both the least financially literate and the least reachable through the digital channels into which financial services are increasingly migrating. A purely digital financial-education campaign would, by construction, miss precisely the people who need it most. This is the kind of actionable insight that the cluster analysis adds beyond the one-variable-at-a-time view.

4.6 Interpretation

Three messages emerge from the results taken together. **First**, financial vulnerability in Italy is *not primarily geographic*: the North–South gap (0.53 points) is small compared with the gaps by work status (3.25), age (1.76) and income (1.62). The popular framing of a simple North–South divide is, for financial literacy, misleading. **Second**, the *youth paradox* indicates that literacy is built through accumulated financial experience rather than through formal education or digital fluency alone — a finding with direct implications for *when* education should be delivered. **Third**, and most importantly, *behaviour beats demographics*: the resilience index alone out-predicts every demographic variable, and behavioural features add fifteen percentage points of predictive accuracy. What people *do* financially, and the buffers they hold, matters more for their literacy than the demographic category they belong to.

5 Conclusion

Gap between the goal and the results. The study set out to identify and characterise financially vulnerable groups in Italy and to understand their drivers. This goal was substantially achieved: the vulnerable groups were clearly located (the unemployed, the young, the low-income and low-education, and a distinct elderly–offline cluster); the gaps were confirmed by formal hypothesis tests; and behaviour was shown, both by correlation and by predictive modelling, to outweigh demographics. The final predictive accuracy of about 70% is moderate rather than spectacular. We regard this honestly as part of the answer, not a shortcoming: it shows that even demographics *and* behaviour together do not fully determine literacy, because important drivers — financial socialisation in the family, exposure to financial decisions, personality — are simply not recorded in the survey. In the language of the course, this is a relevant question, soundly addressed, with a partly "problematic" answer that itself points towards what new data would be needed.

Strengths of the study. A target built on a validated international standard and explicitly checked for internal consistency; a problem-first narrative rather than an algorithm race; convergent evidence from exploratory analysis, significance testing, and both supervised and unsupervised learning; an explicit feature-engineering contribution that both improves the model and, more importantly, changes its interpretation; and transparent treatment of data quality, the bias–variance trade-off, and the sampling weights.

Weaknesses of the study. The cross-sectional design rules out causal claims; the 31% income non-response may bias income-conditioned results; the behavioural items are self-reported; and the macro-regional enrichment rests on only five distinct values, limiting its statistical contribution.

Directions for further research and improvement. Five avenues stand out. (i) Exploit the three IACOFI waves (2017, 2020, 2023) to study how vulnerability evolves over time and across economic shocks. (ii) Apply the survey weights formally throughout the modelling, not only in the descriptive mean. (iii) Model interaction effects explicitly — for example age $\times$ work status — to test whether young unemployed adults are a compounded-risk group. (iv) Develop the new digital-financial-skills items (QS4) to test a "digital double-divide" hypothesis, whereby the same groups that are financially vulnerable are also digitally excluded. (v) Most fundamentally, collect or link richer data on financial socialisation and lived financial experience, which our results suggest are the missing ingredients needed to close the unexplained-variance gap.

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A Reproducibility and Pipeline Detail

The full analysis is organised as five Jupyter notebooks in the public repository `https://github.com/AliSedghiye/DSL_proj`, each corresponding to one stage of the methodology:

1. `01_data_loading.ipynb` — loads the CSV, recodes the negative sentinel values (-97, -98, -99, -999) to missing, inspects missingness, and merges the ISTAT regional enrichment.
2. `02_score_construction.ipynb` — builds the OECD/INFE knowledge, behaviour and attitude subscores and the composite; runs the internal-consistency check and the weighted-versus-unweighted comparison.
3. `03_eda.ipynb` — produces the exploratory figures and the t-test/ANOVA significance results.
4. `04_model.ipynb` — trains the Logistic Regression and Random Forest baselines, plots the confusion matrices and feature importances, and runs the K-Means clustering with the elbow method.
5. `05_feature_engineering.ipynb` — constructs the five behavioural features, retrains Random Forest and Gradient Boosting, and produces the model-comparison and final feature-importance figures.

Modelling details. The full clean dataset has 4,862 rows. For the supervised models, listwise deletion on the selected feature set left 3,026 complete cases; the target (below-median literacy score) was balanced (about 41% positive). Models were evaluated with a stratified 80/20 train–test split and 5-fold cross-validation. The Random Forest used 300 trees with a maximum depth of 8 and balanced class weights; Gradient Boosting used 200 estimators, a maximum depth of 4, and a learning rate of 0.05.

Score summary. Knowledge mean 5.02/7; Behaviour 4.81/9; Attitudes 1.77/5; Composite 11.70/21 (weighted 11.52).